

A Comprehensive Look at Opensource Database Security Concerns

In the computer age, data is wealth, but when stored in databases it is prone to various kinds of vulnerabilities like SQL injections, abuse of the DB's features, broken DBs, data leaks, etc. This article takes a look at the various types of threats to databases and some solutions.



Today, while billions of bytes of data are generated every minute, leading to public cloud computing and the growth of Web 2.0 applications, the relevance of database security becomes all the more paramount. Enterprises are now leveraging open source products for a range of business use cases. Popular open source databases (MySQL, PostgreSQL and MongoDB, to name a few) have seen increased adoption of close to 53 per cent by enterprises, as per statistics. Cost savings, ease of use, avoiding vendor lock-in and community editions are the main reasons for the popularity of open source databases. It is also believed that OSS is more secure because more eyes are looking to find and fix problems. However, security issues need in-depth expertise in this field and not all developers are security experts.

Common attacks to databases and how to address this issue

Databases are an important reservoir of mass information that an enterprise works with. Theft, fraud, loss of privacy, integrity and availability are reasons why data can be compromised. This could be because of deliberate human actions, natural hazards or random events. At the bare minimum, it is required that the database ensures physical and logical integrity. In addition, integrity of each element that makes up the database should be ensured. Access control, user authentication and database availability are other crucial areas that have to be secured. For security hardening, the DBA would need to apply security policies that limit access to data stored on servers.

The common threats that databases face can be categorised as follows.

Excessive privileges: If users are granted more privileges than required, those privileges can be used to gain confidential information. Query-level access control should solve the issue.

Privilege abuse: Users may misuse privileges for unauthorised purposes. Access control policies should be in place to mitigate this.

Unauthorised privilege escalation: Attackers may convert low-level access privileges to high-level privileges.

Platform vulnerabilities: The platform or operating system may be vulnerable to leakage and corruption of data. This could be mitigated by an efficient patch management process and vulnerability assessment activities.

SQL injection: This attack makes a user send unauthorised database queries which lead to the server revealing information. This type of threat is addressed by using prepared SQL statements.

Denial of Service: This involves making the resource unavailable for the purpose it was designed. That is, access to data or the application is denied to the user.

Backup exposure: The backup storage media remains unprotected from any attack.

Database security and open source

Most open source databases have security concerns like access controls and inbuilt secure communication — security

